General Relativity: Weak Fields, Gravitational Waves, and the Action Principle

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Lecture notes organized by LazyingArt LLC with Video2Book

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Chapter 1

Weak Fields, Gravitational Waves, and the Action Principle

The exact equations of general relativity are nonlinear, and their component form can become formidable before one has solved a single physical problem. For the last lecture of the course, it is better to separate principle from bookkeeping. We will ask what Einstein's equations become when the gravitational field is weak, identify the waves contained in that approximation, and then finish by seeing how the full field equations fit inside a remarkably compact action principle.

Three questions organize the discussion. What does a weak gravitational field mean? Why is the weak theory linear although the exact theory is not? And what physical effects distinguish a gravitational wave from a mere wiggle of coordinates?

1.1 A Small Disturbance of an Equilibrium Geometry

Weakness is an approximation scheme. If a dimensionless disturbance is of size $\epsilon \ll 1$, its square is of size ϵ^2, and a first-order calculation consistently discards the latter. The familiar example is a mechanical system making small oscillations about equilibrium. Its exact equations may be nonlinear, but sufficiently small oscillations obey linear equations.

For gravity, the equilibrium background will be empty flat spacetime. Begin with Einstein's vacuum equation,

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0. \tag{1.1}$$

Taking its trace in four dimensions gives

$$g^{\mu\nu} \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) = R - 2R = -R = 0, \tag{1.2}$$

so the vacuum equation is equivalently

$$R_{\mu\nu} = 0. \tag{1.3}$$

Here equilibrium means a source-free solution with no time dependence. Flat spacetime is the simplest such solution. One qualification is essential: flatness does not mean that the components of the metric equal a particular matrix in every coordinate system. It means that inertial coordinates

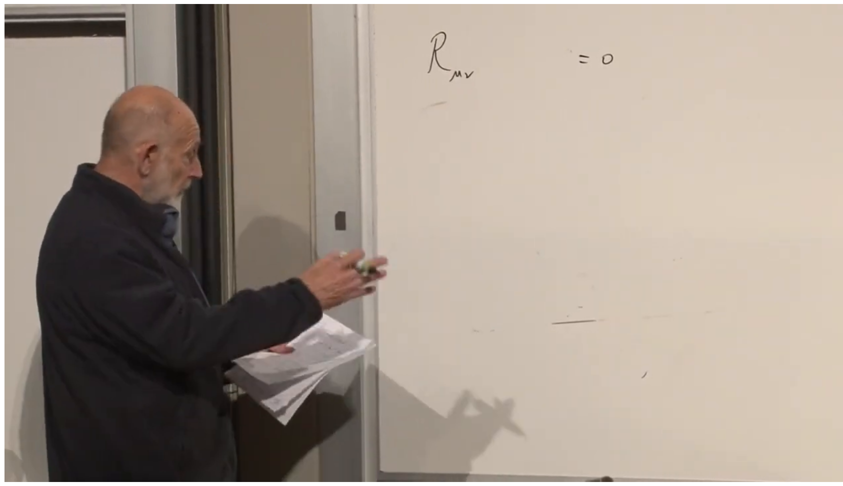

Figure 1.1: The vacuum equation $R_{\mu\nu} = 0$, isolated at the beginning of the weak-field argument.
Lecture frame: 00:03:11.

can be chosen in which

$$g_{\mu\nu} = \eta_{\mu\nu}. \tag{1.4}$$

This distinction between a geometry and the coordinates used to describe it will soon become the central subtlety.

Now imagine a violent source such as a compact binary. Its near field may be strong, but far from the source the disturbance spreads out and becomes small. In a suitable coordinate system the metric can then be written

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \qquad |h_{\mu\nu}| \ll 1. \tag{1.5}$$

The perturbation $h_{\mu\nu}$ is allowed to depend on all four spacetime coordinates. Weakness means that powers beyond the first will be neglected.

1.2 What Survives at First Order

The complete Ricci tensor is too long to be useful at this stage. Its structure is enough. Christoffel symbols contain an inverse metric and one derivative of the metric, while curvature contains a derivative of a Christoffel symbol and a product of two Christoffel symbols:

$$\Gamma \sim \frac{1}{2} g^{-1}\partial g, \qquad R \sim \partial\Gamma + \Gamma\Gamma. \tag{1.6}$$

The inverse metric follows from $g^{\mu\alpha} g_{\alpha\nu} = \delta^{\mu}{}_{\nu}$. To first order,

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + O(h^2), \tag{1.7}$$

where indices on h are raised with the background metric. In inertial coordinates the background is constant,

$$\partial_\alpha \eta_{\mu\nu} = 0, \tag{1.8}$$

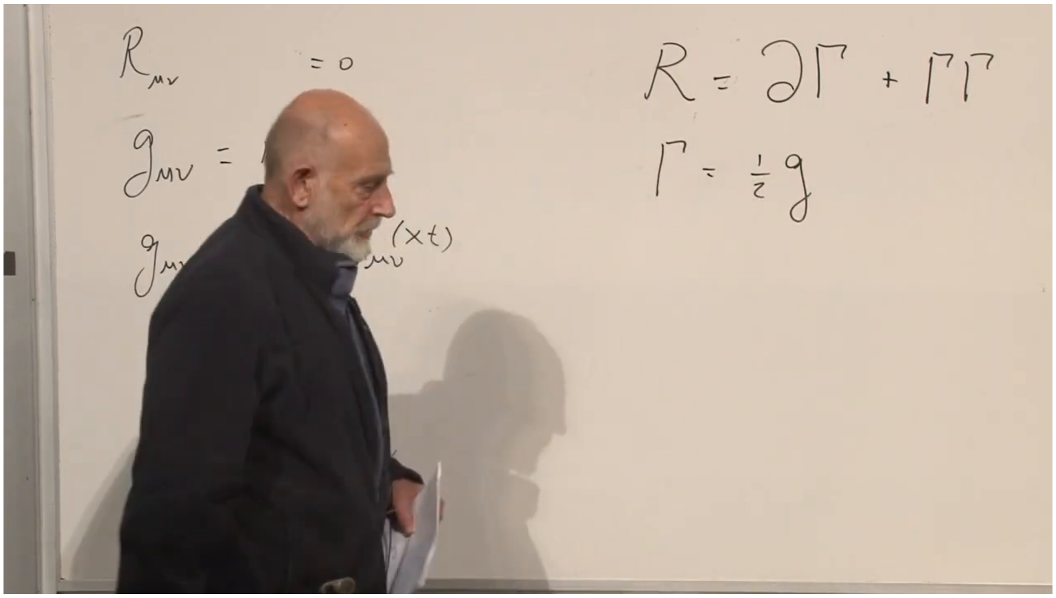

Figure 1.2: The weak-field split and the schematic construction $R \sim \partial\Gamma + \Gamma\Gamma$. The notation exposes the orders in h without expanding every index.

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and therefore

$$\partial_\alpha g_{\mu\nu} = \partial_\alpha h_{\mu\nu}. \tag{1.9}$$

Consequently,

$$\Gamma = O(\partial h), \qquad R_{\mu\nu} = O(\partial\partial h) + O\big((\partial h)^2\big) + O(h\,\partial\partial h). \tag{1.10}$$

The last two terms in Equation (1.10) are quadratic in the disturbance and are omitted. The vacuum equations are therefore linear equations involving second derivatives of $h_{\mu\nu}$. This is the whole meaning of linearization. It is not an assertion that Einstein's exact theory is linear.

Question from the audience. The curvature tensor has four indices, while the Ricci tensor has two. If two derivatives already carry indices and $h_{\mu\nu}$ carries two more, how does the index count work?

Response. The Ricci tensor is obtained by contracting a pair of indices in the curvature tensor. The metric used in that contraction leaves two free indices, exactly the pair carried by $R_{\mu\nu}$. The symbolic form $\partial\partial h$ suppresses those contractions; it is an order count, not a complete indexed formula. [a]

[a] Lecture timestamp: 00:18:45–00:19:53.

Question from the audience. Does contracting with the inverse metric introduce terms such as $h\,\partial^2 h$, and if so, why may they be ignored?

Response. It does. Replacing g^{-1} by $\eta^{-1} - h + \cdots$ produces both the linear term $\eta^{-1}\partial^2 h$ and corrections containing a second h. The latter are quadratic. The same rule discards $h\,\partial h$, $h\,\partial^2 h$, and $(\partial h)^2$. Keeping only one power of the perturbation is precisely the approximation being made. [a]

[a] Lecture timestamp: 00:19:55–00:21:14.

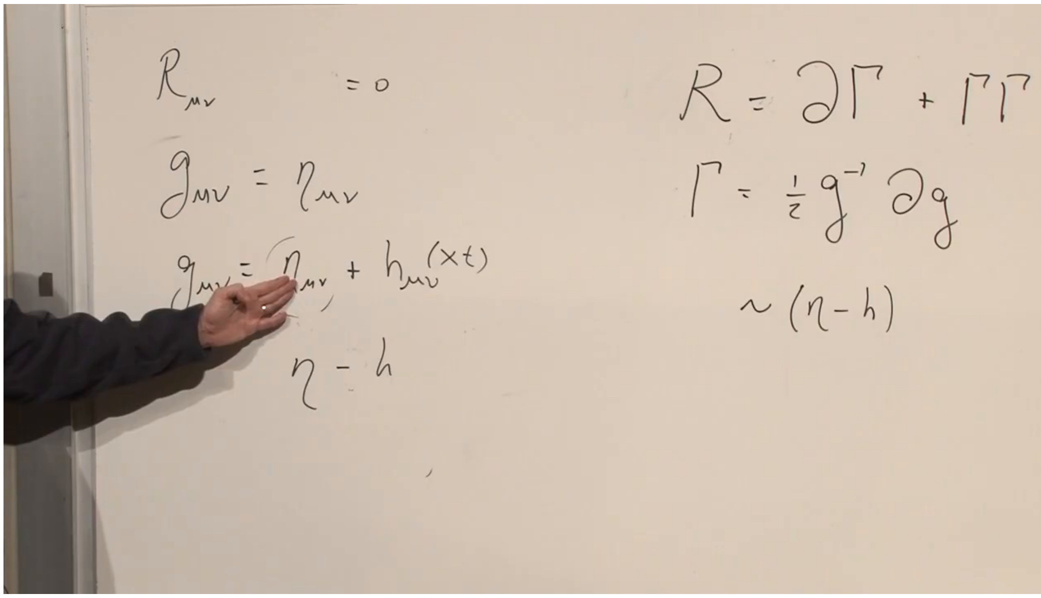

Figure 1.3: The board expansion of the inverse metric and the observation that $\partial\eta = 0$. Only derivatives of $h_{\mu\nu}$ remain at leading order.

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1.3 Why the Equations Look Like Waves

Second-derivative field equations naturally suggest wave propagation. For a scalar function moving along the z-axis, the prototype is

$$\left(\partial_t^2 - \partial_z^2\right)\phi = 0, \tag{1.11}$$

whose solutions include arbitrary right- and left-moving profiles,

$$\phi(z,t) = f(z-t) + g(z+t). \tag{1.12}$$

In three spatial dimensions the same equation is

$$\left(\partial_t^2 - \nabla^2\right)\phi = 0. \tag{1.13}$$

Its linearity permits superposition: the sum of two solutions is another solution.

Gravity is not a scalar field. The perturbation has tensor components, and the equation $R_{\mu\nu} = 0$ initially appears to supply many component equations. They are not all independent; some are constraints and some reflect coordinate freedom. The analogy is with Maxwell theory, where a family of field equations eventually leaves only the physical transverse waves.

After suitable coordinate conditions are imposed, the radiative components obey an ordinary wave equation,

$$\Box h_{\mu\nu} = 0, \qquad \Box = \eta^{\alpha\beta}\partial_\alpha\partial_\beta. \tag{1.14}$$

The precise signs inside $\Box$ depend on the signature convention; its content does not. Disturbances propagate on the light cone.

1.4 Coordinate Wiggles Are Not Gravitational Waves

Before accepting every solution of the linearized equations as a physical wave, we must return to the coordinate caveat. Flat space can be described with curved coordinate lines. Its metric components then differ from their Cartesian values even though its curvature is still zero.

The blackboard itself supplies the example. Draw straight Cartesian coordinates on its flat surface and write

$$ds^2 = \delta_{mn}\, dx'^m dx'^n. \tag{1.15}$$

Now relabel the same points with slightly wiggly coordinates,

$$x'^m = x^m + f^m(x). \tag{1.16}$$

The differentials become

$$dx'^m = dx^m + \partial_n f^m\, dx^n. \tag{1.17}$$

Substituting into the line element and retaining only first order in f gives

$$ds^2 = \delta_{mn} dx^m dx^n + (\partial_m f_n + \partial_n f_m)\, dx^m dx^n + O(f^2). \tag{1.18}$$

Thus an apparent perturbation of the form

$$h_{mn} = \partial_m f_n + \partial_n f_m \tag{1.19}$$

can describe no new geometry at all. It is flat space in wiggly coordinates.

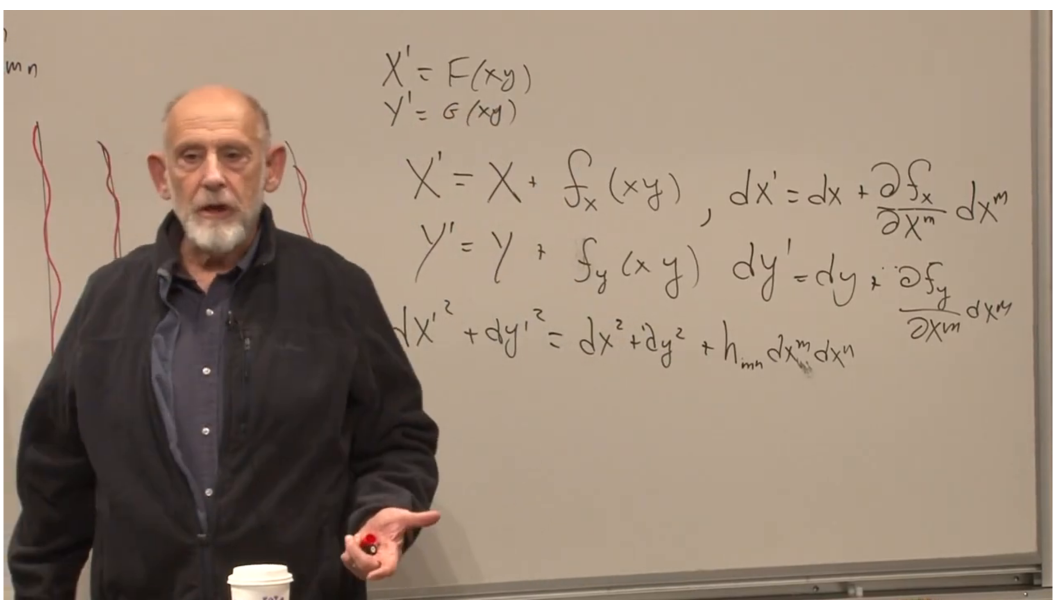

Figure 1.4: The coordinate-wiggle calculation on the board. A small re-coordinatization induces a small-looking h_{mn} even though the blackboard remains flat.

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Equation (1.19) is the linearized form of coordinate, or gauge, freedom. Additional conditions are needed to eliminate these phony perturbations. The detailed gauge algebra is not developed here. Its role is the important point: it removes changes of description while preserving changes of curvature. Once the coordinate freedom and the constraint equations have been accounted for, the physical content reduces to two independent wave polarizations.

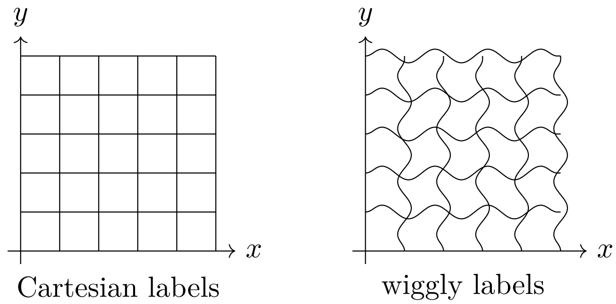

Figure 1.5: The geometry stays flat while the coordinate grid changes. Metric components alone cannot decide whether a perturbation is physical.

1.5 A Wave Moving Along the z-Axis

Choose a plane wave traveling in the positive z-direction. Its phase may be written

$$k(z-t), \tag{1.20}$$

so a representative component has the form

$$h_{\mu\nu}(z,t) = h^{(0)}_{\mu\nu} \sin k(z-t). \tag{1.21}$$

Surfaces of constant phase satisfy $z - t =$ constant; the propagation speed is therefore the speed of light.

In a transverse-traceless description, the components containing a time index or a z-index vanish:

$$h_{0\mu} = 0, \qquad h_{z\mu} = 0. \tag{1.22}$$

Only the x-y block remains. It is symmetric and traceless,

$$h_{xy} = h_{yx}, \qquad h_{xx} + h_{yy} = 0. \tag{1.23}$$

There are consequently two independent patterns. They may be displayed as

$$h^{\mathrm{TT}}_{ij}(z,t) = \begin{pmatrix} h_+ & h_\times \\ h_\times & -h_+ \end{pmatrix} \sin k(z-t), \qquad i,j \in \{x,y\}. \tag{1.24}$$

The diagonal pattern is called the plus polarization. The off-diagonal pattern is called the cross polarization. Rotating the transverse axes by 45° turns one pattern into the other, and an arbitrary linearly polarized wave is a superposition of the two. There are no further independent polarizations in this theory.

1.6 From Metric Components to Physical Strain

Fix z and t and inspect the transverse plane. For a pure plus wave,

$$ds^2_\perp = (1 + h_{xx})\, dx^2 + (1 - h_{xx})\, dy^2. \tag{1.25}$$

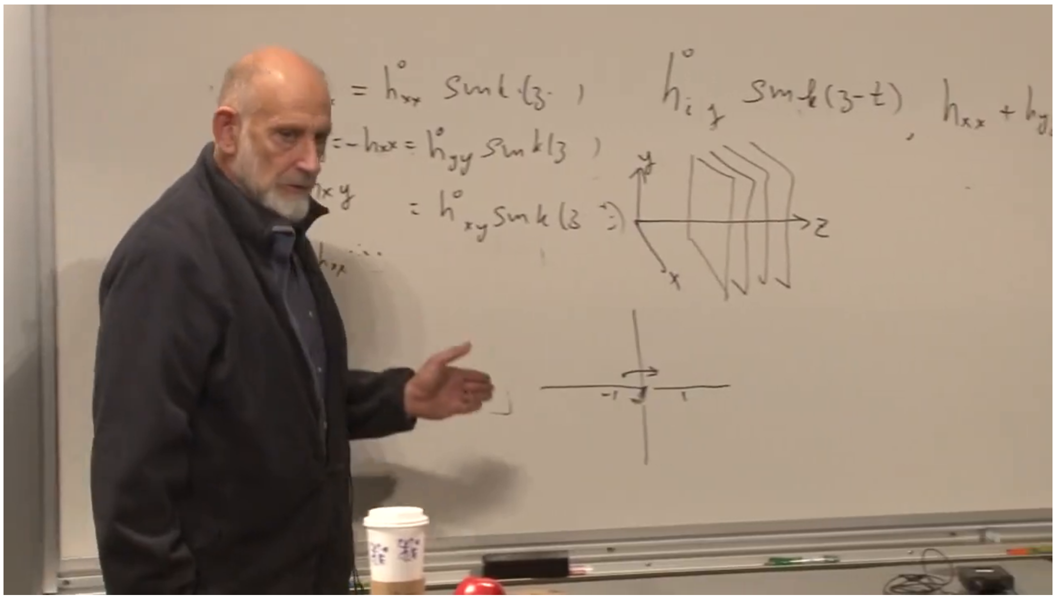

Figure 1.6: A wave traveling along z, represented as a stack of transverse planes. The surviving x-y amplitudes obey $h_{xx} + h_{yy} = 0$.

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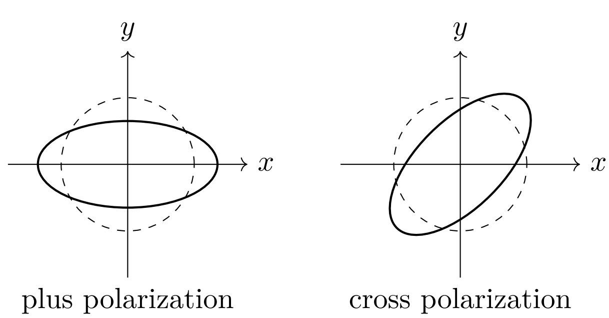

Figure 1.7: One half-cycle stretches one transverse direction while squeezing the perpendicular direction. Half a cycle later the deformation reverses.

When $h_{xx} > 0$, a fixed coordinate interval in the x-direction has a slightly larger proper length, while the corresponding y-interval has a slightly smaller proper length. Half a period later h_{xx} changes sign, and the roles reverse. A square is alternately stretched horizontally and squeezed vertically, then squeezed horizontally and stretched vertically. For the cross polarization, the same alternation occurs along axes rotated by 45°.

To first order, an arm initially of length L changes by

$$\frac{\Delta L}{L} \simeq \frac{1}{2} h_{ij} n^i n^j, \tag{1.26}$$

where $\mathbf{n}$ points along the arm. This formula translates the metric perturbation into the dimensionless strain measured by a detector.

Question from the audience. A meter stick is held together electromagnetically and does not simply change its rest length with the coordinates. Should the picture instead use freely separated test masses?

Response. The distinction is important. A real rod has elasticity, so the tidal field drives stresses inside it; a steel rod and a wooden rod will generally respond by different amounts. Freely suspended test masses avoid that particular elastic response and reveal relative acceleration more directly. In either case the passing wave is not merely relabeling points. Its time-dependent curvature produces a physical effect. [a]

[a] Lecture timestamp: 00:47:23–00:48:48.

The precise local statement is geodesic deviation. For nearby freely falling particles separated by ξ^i,

$$\frac{d^2 \xi^i}{dt^2} = -R^i{}_{0j0}\, \xi^j. \tag{1.27}$$

In the weak transverse-traceless wave,

$$R_{0i0j} = -\frac{1}{2} \ddot{h}^{\mathrm{TT}}_{ij}, \tag{1.28}$$

so the oscillating metric generates alternating tidal acceleration. This is why a time-dependent wave cannot be dismissed as a static choice of rulers. It has nonzero curvature.

Question from the audience. What can measure the squeeze and stretch if every ruler in the region is affected?

Response. A strain gauge can register the genuine stress induced in a material body. Better still, one can monitor the changing separation of freely suspended masses. A static, uniform rescaling could be absorbed into the measuring apparatus, but the oscillating curvature drives a response that can be recorded. [a]

[a] Lecture timestamp: 00:51:10–00:52:02.

1.7 Sources and Detectors

Electromagnetic waves are generated by moving charges and currents. Gravitational waves are generated by moving distributions of mass and energy. Conservation of energy and momentum rules out gravitational monopole and dipole radiation for an isolated system, so the leading radiative moment is quadrupolar. A compact binary supplies exactly the needed changing quadrupole: two dense bodies orbit one another and launch waves into the far field.

Close to the binary, spacetime is strongly curved and the linear expansion of Equation (1.5) is unreliable. Far away, the outgoing wave spreads over a larger area, its amplitude becomes small, and the linearized description becomes accurate. For an observer on the z-axis from the source, the measurable stretching lies in the transverse x-y plane.

Question from the audience. Is the ruler in the drawing an abstract coordinate ruler or a physical meter stick, and how could a variable ruler measure the effect?

Response. It may be a physical object, but then its material properties matter. Curvature supplies the tidal stress; steel and wood need not deform identically under it. Comparing their responses is already a measurement. A cleaner detector replaces the rods by separated test masses and measures their differential motion. [a]

[a]Lecture timestamp: 00:54:00–00:55:19.

A resonant detector is one strategy. If a mechanical system has a natural frequency near that of the incoming wave, the wave drives it coherently and amplifies the response. An interferometer uses another strategy: mirrors serve as test masses, and laser interference compares the effective lengths of perpendicular arms. A plus-polarized wave lengthens one arm while shortening the other, which is exactly the differential signal an interferometer is designed to detect.

Question from the audience. Would this be the basis of a gravitational-wave detector?

Response. Yes. A stiff rod barely changes, whereas freely suspended end masses can undergo relative acceleration. One can also exploit resonance so that a weak periodic drive produces a larger response. The expected astrophysical strain is extraordinarily small, of order 10^{-21}, so detector design is the art of turning that tiny differential motion into a measurable signal. [a]

[a]Lecture timestamp: 00:55:20–00:57:10.

Question from the audience. Is LIGO an example of such a detector?

Response. Yes. Its basic observables come from laser light reflected between mirrors in perpendicular arms. A gravitational wave changes the relative optical path lengths and therefore the interference pattern. At the time of this lecture the instrument had not yet made a direct detection; that historical situation later changed.[a] [b]

[a]Editorial clarification: The first direct observation, GW150914, was recorded in 2015 and announced in 2016.
[b]Lecture timestamp: 00:57:16–00:58:05.

Question from the audience. What propagation speed follows from the wave solution?

Response. The speed of light. The phase $k(z-t)$ moves one unit in z for one unit in t, and the linearized Einstein equation has the same light-cone propagation structure as the electromagnetic wave equation. [a]

[a]Lecture timestamp: 00:58:07–00:58:16.

Black-hole collisions and compact binaries are especially promising sources. Near the collision the gravitational field is violently nonlinear, and a substantial part of the released energy may leave as gravitational radiation. By the time the signal reaches a distant detector, its amplitude is weak enough for linear wave propagation to be an excellent description. Gravitational radiation thus opens an astronomical channel that need not depend on emitted light.

1.8 The Binary Pulsar and Indirect Evidence

Radiation carries energy. If a binary emits gravitational waves, the orbital energy decreases, the bodies spiral slowly inward, and the orbital period changes. The binary pulsar provides a clock precise enough to measure that change. Its observed orbital evolution agrees quantitatively with the energy loss predicted by general relativity.

In the historical setting of the lecture, this was the decisive indirect evidence: no detector had yet registered a wave passing through Earth, but the orbit was losing energy at the rate demanded by gravitational radiation. The waves were already observable through what they did to their source.

Question from the audience. What frequency or period characterizes the original binary-pulsar system? Is it on the millisecond scale?

Response. The millisecond scale refers to a pulsar's spin, not to the orbit of the two stars. The classroom checks the value and finds an orbital period of about 7.75 hours. The distinction matters: the gravitational radiation tracks the binary orbital motion, while the radio pulses supply the precise clock used to follow it. [a]

[a]Lecture timestamp: 01:01:27–01:03:28.

Question from the audience. How much did the accumulated orbital timing change over the decades of observation?

Response. The quoted accumulated shift is roughly thirty seconds. The instantaneous fractional change is tiny, but decades of precise pulse timing make the cumulative effect visible. [a]

[a]Lecture timestamp: 01:03:34–01:03:54.

1.9 Time Dependence, Curvature, and the Limits of Linearity

Question from the audience. Why choose the transverse x-y plane? What happens if one instead considers a plane involving x and t?

Response. The wave itself depends on both z and t, but transverse refers to the spatial directions perpendicular to its spatial direction of propagation. For a wave moving along z, the physical metric amplitudes may be chosen in the x-y block. Other apparent components are absent or removable coordinate wiggles. [a]

[a]Lecture timestamp: 01:02:22–01:03:17.

Question from the audience. Is time itself oscillating?

Response. No. The wave oscillates as a function of time. At a fixed spatial position, the components $h_{ij}(z,t)$ vary periodically. Time is a coordinate on which the field depends, not an object being alternately stretched and compressed in this picture. [a]

[a]Lecture timestamp: 01:04:00–01:04:18.

Question from the audience. If all $h_{0\mu}$ vanish, is there no curvature involving the time direction?

Response. There is. The surviving spatial components depend on time, and curvature contains two derivatives of them. Thus derivatives such as $\partial_t^2 h_{ij}$ generate components R_{0i0j}. Vanishing time-index components of $h_{\mu\nu}$ do not imply vanishing time-dependent curvature. [a]

[a] Lecture timestamp: 01:05:15–01:06:09.

The linear wave equation arose because h was assumed small, not because coordinate artifacts were removed. These are two separate operations. Gauge fixing identifies the physical degrees of freedom; linearization discards nonlinear interactions.

Question from the audience. Einstein's equations are nonlinear, but the resulting wave equation is linear. Is that linearity caused by eliminating nonphysical solutions?

Response. No. It is caused by the small-amplitude approximation. Terms containing two powers of h were thrown away. Removing coordinate wiggles decides which solutions are physical; it is not what makes the equation linear. [a]

[a] Lecture timestamp: 01:06:45–01:07:29.

Question from the audience. After solving the linearized equation, how large is the error compared with an exact solution?

Response. The first omitted corrections are of order h^2. If $h \sim 10^{-1}$, they are of order 10^{-2}; if $h \sim 10^{-3}$, they are of order 10^{-6}. A perturbative calculation must always be checked afterward to ensure that the assumed small parameter was actually small. [a]

[a] Lecture timestamp: 01:07:35–01:08:31.

Question from the audience. When higher-order terms become important, does the wave lose its transverse character?

Response. The essential transverse radiative character remains, but linear superposition does not. Strong waves interact with the gravitational field carried by other waves; instead of passing through one another unchanged, they scatter. [a]

[a] Lecture timestamp: 01:08:34–01:09:11.

Question from the audience. A mechanical wave speed depends on stiffness and density. Is there a comparable sense in which spacetime has a stiffness?

Response. Only as an analogy. For a material medium, propagation depends on a combination such as stiffness divided by density. Since electromagnetic and gravitational waves propagate at c, the old mechanical language would call their medium extraordinarily stiff. Modern physics does not assign spacetime an ordinary Young's modulus; the invariant fact is the light-speed propagation fixed by the field equations. [a]

[a] Lecture timestamp: 01:09:12–01:11:10.

1.10 A Second Route: Stationary Action

With the wave discussion complete, there is time for one final change of viewpoint. The principle of stationary action is the common language of classical mechanics, electromagnetism, field theory, and the Standard Model. General relativity belongs to the same structure.

For a particle, the action is an integral along a trajectory. For a field, a history assigns a field value $\phi(x)$ at every point of spacetime, and the action is an integral over spacetime:

$$S[\phi] = \int d^4x \, \mathcal{L}(\phi, \partial_\mu \phi). \tag{1.29}$$

Every sufficiently regular $\phi(x)$ is a conceivable history, just as every sufficiently regular path is a conceivable particle trajectory, but most are not solutions. The physical histories are stationary points of the action subject to fixed boundary data:

$$\delta S = 0. \tag{1.30}$$

For a field, the boundary surrounds a spacetime region rather than merely specifying the two endpoints of a path. The Euler–Lagrange equations then select the interpolation between the prescribed boundary values.

Einstein's field equations can be obtained in exactly this way. The variation is long, but the action itself is simple. Before writing it, we need the proper measure of spacetime volume.

1.11 Why the Determinant Measures Volume

Begin in two dimensions with a diagonal metric,

$$ds^2 = g_{xx}\, dx^2 + g_{yy}\, dy^2. \tag{1.31}$$

A coordinate rectangle of sides dx and dy has proper side lengths $\sqrt{g_{xx}}\, dx$ and $\sqrt{g_{yy}}\, dy$, so its proper area is

$$dA = \sqrt{g_{xx} g_{yy}}\, dx\, dy. \tag{1.32}$$

If the off-diagonal component g_{xy} is present, the coordinate edges need not be orthogonal. The corrected area is

$$dA = \sqrt{g_{xx} g_{yy} - g_{xy}^2}\, dx\, dy = \sqrt{\det g}\, dx\, dy. \tag{1.33}$$

The same construction works in any dimension:

$$dV_n = \sqrt{|\det g|}\, d^n x. \tag{1.34}$$

For Lorentzian spacetime, if $g = \det(g_{\mu\nu})$, the invariant four-volume element is conventionally written

$$dV_4 = \sqrt{|g|}\, d^4x, \tag{1.35}$$

or $\sqrt{-g}\, d^4x$ in the mostly-plus convention. The coordinates of a region may change, but its physical four-volume does not.

Multiplying this measure by any scalar $F(x)$ and integrating produces a coordinate-invariant quantity:

$$\int d^4x \sqrt{|g|}\, F(x). \tag{1.36}$$

An action for a generally covariant theory should have this form. Otherwise the field equations derived from it could depend on an arbitrary choice of coordinates.

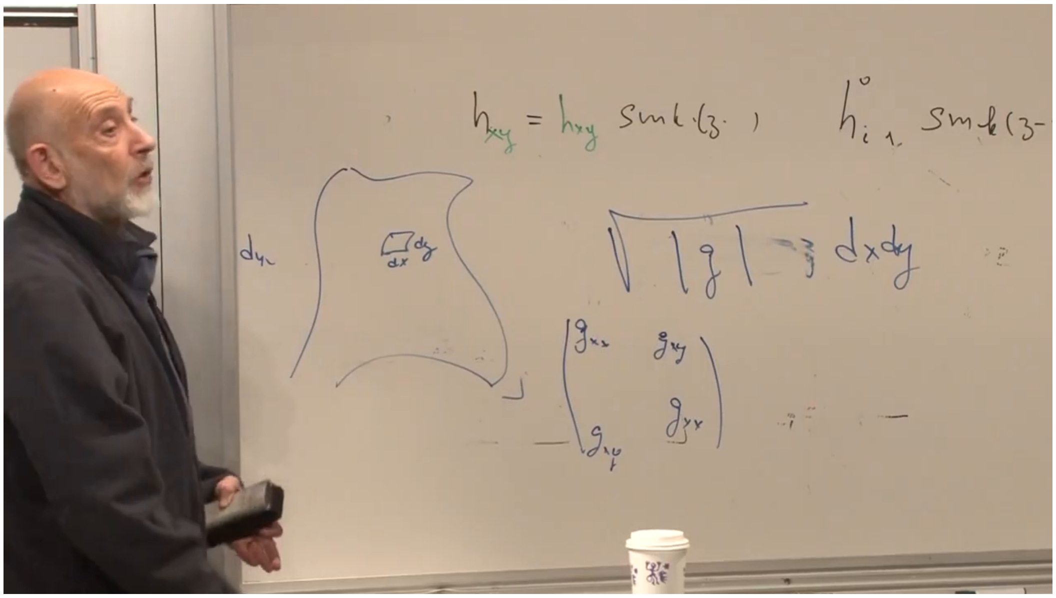

Figure 1.8: The board generalization from a small coordinate rectangle to the proper area element $\sqrt{|g|}\,dx\,dy$. The determinant incorporates both diagonal lengths and off-diagonal shear.

Lecture frame: 01:21:25.

1.12 The Einstein–Hilbert Action

For pure gravity, the dynamical field is the metric itself. The simplest nonconstant scalar made from the metric and its first two derivatives is the scalar curvature R. A constant is also a scalar; its coefficient is the cosmological constant. Up to sign conventions and boundary terms, a standard normalization of the gravitational action is

$$S_{\text{grav}} = \frac{1}{16\pi G}\int d^4x\,\sqrt{|g|}\,(R - 2\Lambda). \tag{1.37}$$

The term proportional to Λ is simply a multiple of total spacetime volume. The earlier wave equation was developed for vacuum with $\Lambda = 0$, so that term did not appear there.

Matter is added through a matter action,

$$S = S_{\text{grav}} + S_{\text{matter}}. \tag{1.38}$$

For electromagnetism, for example, the invariant field contribution is built from the antisymmetric tensor $F_{\mu\nu}$:

$$S_{\text{EM}} = -\frac{1}{4}\int d^4x\,\sqrt{|g|}\,F_{\mu\nu}F^{\mu\nu}. \tag{1.39}$$

Varying the total action with respect to the inverse metric defines the stress-energy tensor and gives

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G\,T_{\mu\nu}. \tag{1.40}$$

Thus the compact action contains the complete field equations. In vacuum, with $\Lambda = 0$, it reduces to the theory that governed the gravitational waves studied earlier.

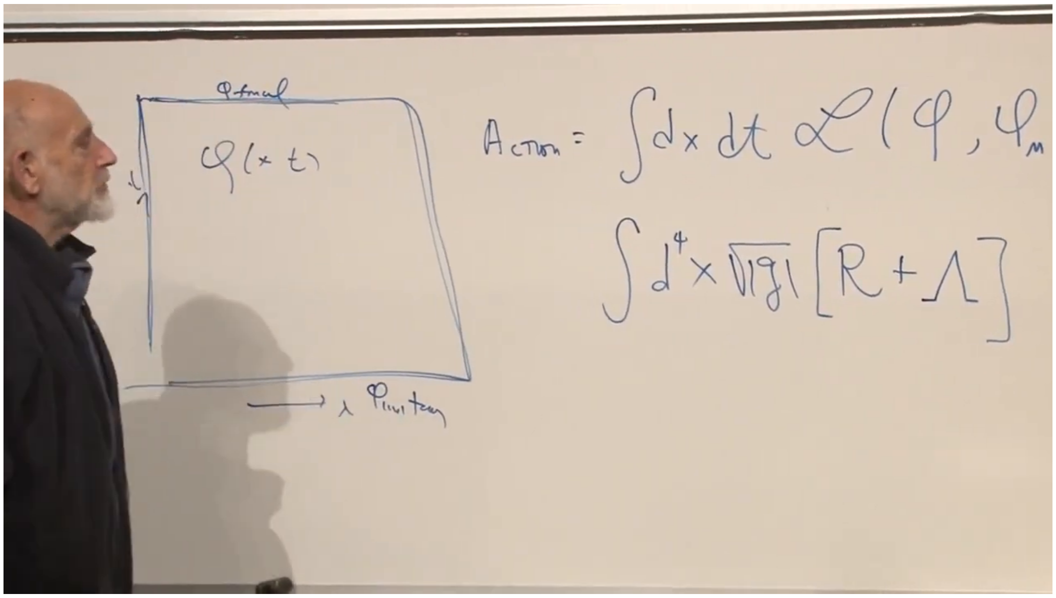

Figure 1.9: The field-history picture and the gravitational action on the board. The scalar curvature supplies the dynamics; the constant term is proportional to spacetime volume.

Lecture frame: 01:26:32.

Question from the audience. Where does the stress-energy tensor enter the action principle?

Response. It comes from the additional terms describing matter and other fields. Electromagnetic energy, particle fields, and any other material content contribute to S_{matter}. Their metric variation supplies $T_{\mu\nu}$. The integral containing only $\sqrt{|g|}R$ is the vacuum gravitational theory. [a]

[a] Lecture timestamp: 01:27:26–01:28:25.

The derivation is conceptually straightforward and algebraically unpleasant. One constructs the Christoffel symbols from the metric, constructs the curvature, contracts to form R, multiplies by the determinant factor, and varies the resulting expression. The lecture jokes that the exercise should take five minutes and that failure would mean failing the course, then immediately admits to having abandoned the calculation many times after much longer attempts. The joke makes a serious point: compact principles need not imply short component calculations.

Question from the audience. Is the complete variation normally written out in a relativity text?

Response. It can be, but it fills pages. A long calculation becomes illuminating only when one works through enough of it to understand why the next step follows; passively reading line after line rarely produces that understanding. Modern symbolic and numerical tools can perform much of the bookkeeping, but they do not replace the organizing principle. [a]

[a] Lecture timestamp: 01:31:10–01:34:16.

There are two traditional ways through such calculations. One may develop a geometric intuition strong enough to see which tensor combinations are possible, or one may push the algebra carefully. Most work uses some mixture of both, often with a computer handling the longest contractions. The course has deliberately emphasized the principles that survive whichever route is chosen.

Question from the audience. In hindsight the action looks highly motivated. Why did the path from special relativity to general relativity take Einstein so many years, and was he aiming toward the final theory from early on?

Response. The endpoint looks simpler after the necessary geometry is known. At the time, the needed differential geometry was unfamiliar to most physicists, and Einstein had to learn and adapt it with mathematical help, notably from Marcel Grossmann. Building the concepts of curved spacetime and finding the correct field equation in roughly a decade was not a slow achievement. The early and final papers are worth reading precisely because they show how much had to be invented along the way. [a]

[a] Lecture timestamp: 01:34:20–01:35:59.

The quarter therefore ends with two complementary forms of the same theory. The field equation tells curvature how to respond locally to stress-energy. The action principle says that the entire spacetime history is a stationary point of one invariant functional. In the weak far field, both descriptions reduce to transverse waves carrying real curvature at the speed of light.